Appendix 3: Data Collection Forms

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

This appendix contains every form used to collect data in the study. Forms 3B-3E, which clinicians complete or take part in, are also provided in Sinhala and Tamil (Appendix 3, Sinhala and Tamil versions); the clinician uses the language they prefer. Form 3A is in English only, because it is completed by hospital staff from English-language medical records and is never given to patients. No form records a patient's name, NIC number, hospital/BHT or clinic number, address, telephone number or any full date. Clinicians are identified only by a reader code.

Form	Purpose	Completed by	When
3A Case Report Form (CRF)	Abstract routinely recorded clinical, imaging, pathology, treatment and follow-up data for the de-identified registry	Designated hospital staff / trained abstractor under the clinical co-supervisor	Once per patient, updated at each follow-up
3B Clinician Background Form	Describe the reader panel (aggregate reporting only)	Participating clinician	Once, after consent
3C Reader-Study Case Record	Record each reader's decision on each case, with and without AI support	Participating clinician (time is also logged by the study software)	Every case, every session
3D Post-Session Questionnaire	System Usability Scale, NASA Task Load Index and trust questionnaire	Participating clinician	End of each reading session
3E Interview Guide	Semi-structured interview on trust, escalation, workload and use in practice	Principal Investigator (interviewer)	Once, optional, separate consent

Appendix 3A: Case Report Form (CRF)

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Instructions. Complete from the hospital record. Write the study code only; do not write any identifier on this form. Record time as the number of days from the first diagnostic MRI (Day 0). Tick one box unless stated otherwise. Write NR if not recorded. Completed CRFs are kept at the hospital and entered into the registry under the study code.

Section 1. Administration

Study code	BT-____-_____	Site code	_____
Data source	☐ Retrospective record (waiver) ☐ Prospective (consent)	Consent given by	☐ Patient ☐ Parent/guardian ☐ Legal representative ☐ N/A (waiver)
Abstractor code	_____	CRF entry (Day)	_____

Section 2. Patient characteristics

Age at diagnosis (years)	_____	Sex	☐ Male ☐ Female
Age group	☐ <18 ☐ 18-39 ☐ 40-59 ☐ 60+	Handedness	☐ Right ☐ Left ☐ NR
Relevant history	☐ Previous cancer ☐ Previous cranial radiotherapy ☐ Genetic syndrome ☐ None ☐ NR		

Section 3. Presentation

Presenting features (tick all)	☐ Headache ☐ Seizure ☐ Focal weakness ☐ Speech disturbance ☐ Visual disturbance ☐ Cognitive / behavioural change ☐ Raised ICP features ☐ Incidental ☐ Other: _____
Duration of symptoms (weeks)	_____
Performance status at presentation	KPS _____ / Lansky (child) _____ / ECOG _____ GCS _____ /15

Section 4. Imaging (one row per MRI study; Day 0 = first diagnostic MRI)

Study	Day	Scanner make / field strength	Sequences available (tick)	Quality	De-identified & defaced
Pre-op	0	_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐SWI ☐Perfusion ☐DTI ☐MRS	☐Good ☐Usable ☐Poor	☐Yes ☐No
Post-op		_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐Other	☐Good ☐Usable ☐Poor	☐Yes ☐No
Other		_____ / ___ T	☐T1 ☐T1c ☐T2 ☐FLAIR ☐DWI ☐Other	☐Good ☐Usable ☐Poor	☐Yes ☐No

Lesion location (from report)	☐ Frontal ☐ Temporal ☐ Parietal ☐ Occipital ☐ Insular ☐ Deep/thalamic ☐ Ventricular ☐ Sellar/suprasellar ☐ Cerebellar ☐ Brainstem ☐ Extra-axial ☐

	Spinal
Side / number of lesions	☐ Left ☐ Right ☐ Bilateral ☐ Midline Number of lesions: _____
Radiological impression	_____

Section 5. Surgery and tissue

Procedure	☐ Stereotactic / needle biopsy ☐ Open biopsy ☐ Subtotal resection ☐ Gross total resection ☐ No surgery
Day of procedure	_____
Extent of resection (post-op MRI)	☐ Gross total ☐ Near total ☐ Subtotal ☐ Partial ☐ Biopsy only ☐ NR
Peri-operative complication	☐ None ☐ New neurological deficit ☐ Haemorrhage ☐ Infection ☐ Other: _____
Archived slides	Number of H&E slides _____ Number of IHC slides _____ Digitised: ☐ Yes ☐ No Slide quality: ☐ Good ☐ Usable ☐ Poor

Section 6. Pathology and reference diagnosis

Histological diagnosis (report)	_____
Integrated diagnosis (WHO CNS5, 2021)	_____
CNS WHO grade	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ Not applicable ☐ NR
Markers (where tested)	IDH ☐Mut ☐WT ☐NT 1p/19q ☐Codel ☐Intact ☐NT MGMT ☐Methyl ☐Unmeth ☐NT ATRX ☐Lost ☐Retained ☐NT p53 ☐Pos ☐Neg ☐NT H3 K27M ☐Pos ☐Neg ☐NT Ki-67 _____ %
Reference standard confirmed by	☐ Consultant histopathologist ☐ Multidisciplinary team ☐ Imaging + clinical only (no tissue)

Section 7. Treatment

Radiotherapy	☐ Yes ☐ No ☐ NR Day started _____ Dose _____ Gy / _____ fractions
Chemotherapy	☐ Temozolomide ☐ Other: _____ ☐ None ☐ NR
Other treatment	☐ Steroids ☐ Anti-epileptic drugs ☐ Shunt / drain ☐ Palliative care only ☐ Other: _____

Section 8. Follow-up (one row per visit)

Visit	Day	MRI done	Response (RANO)	Clinical status	KPS
1		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
2		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
3		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	
4		☐Y ☐N	☐CR ☐PR ☐SD ☐PD ☐NA	☐Stable ☐Improved ☐Worse	

Status at last contact	☐ Alive, no progression ☐ Alive with progression ☐ Died ☐ Lost to follow-up Day of last contact / death: _____
Recurrence	☐ No ☐ Yes Day _____

Section 9. Completeness and sign-off

Data available	☐ MRI ☐ Digitised slides ☐ Markers ☐ Treatment ☐ Follow-up
Identifier check	☐ I confirm that no name, NIC, hospital/BHT/clinic number, address, telephone number or full date appears on this form or in the exported data.
Abstractor code / signature / date	
Checked by (clinical co-supervisor or delegate)	

Appendix 3B: Clinician Background Form

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Please complete once, after signing the consent form. Your answers are stored under your reader code and are reported only as group totals.

Reader code	R-____
Specialty	☐ Neurosurgery ☐ Radiology ☐ Histopathology ☐ Other: _____
Current grade	☐ Consultant ☐ Senior registrar ☐ Registrar ☐ Other: _____
Years in specialty (since starting specialty training)	☐ <5 ☐ 5-9 ☐ 10-19 ☐ 20+
Brain-tumour cases you see or report per month (approx.)	☐ <5 ☐ 5-14 ☐ 15-29 ☐ 30+
Previous use of AI tools in clinical work	☐ Never ☐ Tried once or twice ☐ Occasionally ☐ Regularly
General attitude to AI in your specialty	☐ Very negative ☐ Negative ☐ Neutral ☐ Positive ☐ Very positive

Appendix 3C: Reader-Study Case Record

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

How the reader study works. Each clinician reviews the same set of de-identified cases twice: once without AI support and once with AI support, in separate sessions at least four weeks apart (washout). Half the readers start with the unassisted condition and half with the assisted condition. Case order is randomised within each session. Reading time is recorded automatically by the study software; this record captures the reader's decision. One row per case.

Reader code	R-____	Session no.	____	Condition	☐ Unassisted ☐ AI-assisted

Columns: Dx = most likely tumour type (WHO CNS5 family); Gr = CNS WHO grade 1-4; Conf = your confidence 1 (very unsure) to 5 (very sure). AI-assisted sessions only: AI = what you did with the AI output (A = accepted, M = amended, R = rejected); Esc = for cases the AI referred to you, was referral appropriate? (Y/N); for cases it did not refer, should it have? (Y/N).

#	Case ID	Dx (type)	Gr	Conf	AI (A/M/R)	Esc (Y/N)	Comment / reason for change
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Appendix 3D: Post-Session Questionnaire

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Reader code R-____ Session ____ Condition: ☐ Unassisted ☐ AI-assisted Date _____

Part 1 (workload) is completed after every session. Parts 2 and 3 are completed after AI-assisted sessions only. About 10-15 minutes.

Part 1. NASA Task Load Index (raw TLX)

Hart & Staveland (1988). For each scale, mark a cross on one of the 21 positions to show how the session you have just finished felt.

Scale	Question	Very low ← → Very high
Mental demand	How mentally demanding was the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Physical demand	How physically demanding was the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Temporal demand	How hurried or rushed was the pace of the task?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Performance	How successful were you in accomplishing what you were asked to do? (left = perfect, right = failure)	Perfect Failure |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Effort	How hard did you have to work to accomplish your level of performance?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|
Frustration	How insecure, discouraged, irritated, stressed and annoyed were you?	0 . . . . . 100 |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|

Part 2. System Usability Scale (SUS)

Brooke (1996). Tick one box per statement. 1 = strongly disagree, 5 = strongly agree. "System" means the AI decision-support system you used in this session.

#	Statement	1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Part 3. Trust in the AI decision-support system

Items adapted from Jian, Bisantz & Drury (2000) and written for this study. 1 = strongly disagree, 5 = strongly agree.

#	Statement	1	2	3	4	5
1	I could rely on the system's outputs when it did not refer a case to me.	☐	☐	☐	☐	☐
2	The confidence the system reported matched how correct its outputs were.	☐	☐	☐	☐	☐
3	The system referred the right cases to me for full review.	☐	☐	☐	☐	☐
4	The system missed cases that I think it should have referred to me.	☐	☐	☐	☐	☐
5	The explanations (outlines, heat-maps, evidence breakdown) helped me judge whether the output was correct.	☐	☐	☐	☐	☐
6	I was suspicious of the system's outputs.	☐	☐	☐	☐	☐
7	Using the system did not reduce my own responsibility for the decision.	☐	☐	☐	☐	☐
8	I would be comfortable using this system as a second reader in routine practice.	☐	☐	☐	☐	☐
9	The system could lead a less experienced clinician into error.	☐	☐	☐	☐	☐
10	Overall, I trust this system.	☐	☐	☐	☐	☐

Any comments on this session: _____

_____

Appendix 3E: Semi-Structured Interview Guide

Version 1, 09.11.2026

Study: Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Optional interview with clinicians who completed the reader study and gave separate consent for interview and audio recording. About 30-45 minutes, in person or online, at a time chosen by the clinician, in Sinhala, Tamil or English as the clinician prefers. The questions below are a guide; follow-up questions depend on the answers. The interviewer does not comment on the clinician's own reading performance.

Opening (about 3 minutes)

- Thank the participant; confirm they have read the information sheet and signed the interview consent item.
- Confirm consent to audio recording and start the recorder only after verbal agreement. Remind them that they can skip any question or stop at any time.
- Explain that quotations will be anonymised and that the recording will be deleted after transcription.

1. Current practice

- Can you walk me through how a new brain-tumour case reaches you and how you reach a decision?
- Where in that process do delays or uncertainty usually occur?

2. Experience of the system

- What was it like to review cases with the AI support compared with without it?
- Which outputs did you find most and least useful (outlines, type and grade, evidence breakdown, surgical route)?

3. Confidence and referral (escalation)

- How did you interpret the confidence the system showed?
- Did it refer the right cases to you? Can you recall a case where it referred something it should not have, or did not refer something it should have?
- Where would you set the line between quick sign-off and full review?

4. Trust and responsibility

- What made you trust or distrust a particular output?
- Who do you feel is responsible for a decision made with AI support?
- How do you think the system would affect less experienced clinicians?

5. Workload

- Did the system change how much time or effort the cases took? In what way?
- Were there cases where checking the AI added work?

6. Surgical-planning proof of concept (neurosurgeons)

- How useful was the 3D model and candidate route for thinking about an approach?
- What would it need to show before you would consider it in planning?

7. Use in practice in Sri Lanka

- What would stop or help this kind of system being used in your unit?
- What safeguards would you want in place (training, audit, governance, data protection)?

Closing (about 2 minutes)

- Is there anything we have not discussed that you think is important?
- Stop the recording. Thank the participant and explain how they can obtain the summary of findings.